

Suggested Solutions to Exercises 9

1. Following the usual argument for risk-neutral pricing in continuous time, in a risk-neutral world the *expected* rate of return must be r . Therefore the geometric Brownian motion that a stock will follow will be

$$dS_t = rS_t dt + \sigma S_t dB_t,$$

so $\log S_T$ will follow a normal distribution, with

$$(\log S_T - \log S - (r - \frac{\sigma^2}{2})T) / \sigma\sqrt{T} \sim \mathcal{N}(0, 1),$$

where S is the value of the asset today.

Writing a standard normal variable as Z and rearranging, we obtain

$$\log S_T = \log S + (r - \frac{\sigma^2}{2})T + \sigma\sqrt{T}Z.$$

We now have that

$$\begin{aligned} S_T \geq K &\iff \log(S_T) \geq \log(K) \iff \\ Z &\geq -\frac{\log(S/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = -d_2 \end{aligned}$$

The payoff of the option is 1 when S_T is above the strike K , and 0 otherwise. The risk-neutral expected payoff is therefore

$$\mathbb{P}[S_T \geq K] = \mathbb{P}[Z \geq -d_2] = \mathbb{P}[Z \leq d_2] = N(d_2)$$

where $N(\cdot)$ is the cdf of the standard normal distribution. The value of the digital call is this risk-neutral expected payoff, discounted at the risk free rate, which gives

$$C_{dig} = e^{-rT} N(d_2).$$

2. Let's wait for next week for this.
3. Consider the probability that S_T is below K_1 , where we now work with all probabilities and expectations in the risk-neutral world.

$$\begin{aligned} P[S_T < K_1] &= P[\log(S_T) < \log(K_1)] \\ &= P[\log(S_T) - \log(S_0) < \log(K_1) - \log(S_0)] \end{aligned}$$

We know that $[\log(S_T) - \log(S_0)] \sim N[(r - \sigma^2/2)T, \sigma^2 T]$. Then

$$P[S_T < K_1] = P\left[Z < \frac{\log(K_1) - \log(S_0) - (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right]$$

where $Z \sim N(0, 1)$. This is equal to

$$\begin{aligned} P[S_T < K_1] &= P\left[Z > \frac{\log(S_0/K_1) + (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right] \\ &= P[Z > d_2(K_1)] = N(-d_2(K_1)) = 1 - N(d_2(K_1)), \end{aligned}$$

where $N(\cdot)$ is the cdf of the standard normal distribution, and

$$d_2(K_1) = \frac{\log(S_0/K_1) + (r - \sigma^2/2)T}{\sigma\sqrt{T}}$$

is the usual constant defined in the Black-Scholes formula, for strike price K_1 . Similarly, $P[S_T > K_2] = N(d_2(K_2))$. The derivative payoff function is:

$$\text{Payoff} = \begin{cases} H_1, & \text{if } S_T < K_1; \\ H_2, & \text{if } S_T > K_2; \\ 0, & \text{otherwise.} \end{cases}$$

The expected risk-neutral payoff can then be computed as:

$$\begin{aligned} \text{Expected payoff} &= H_1 \times P[S_T < K_1] + H_2 \times P[S_T > K_2] \\ &= H_1(1 - N(d_2(K_1))) + H_2 N(d_2(K_2)). \end{aligned}$$

The risk-neutral valuation says that the value of the option is the discounted expected risk-neutral payoff, hence

$$f = e^{-rT} [H_1 \{1 - N(d_2(K_1))\} + H_2 N(d_2(K_2))].$$

4. The price of the derivative can be calculated as

$$e^{-rT} \hat{E} [S_T^2]$$

where $\hat{E}$ denotes expectation under the risk-neutral probability distribution. Under the risk-neutral probability distribution, $\log S_T$ follows a normal distribution with mean $\log S_0 + (r - \frac{1}{2}\sigma^2)T$ and variance $\sigma^2 T$. Using the inverse cdf method on uniform $(0, 1)$ variables we know that $N^{-1}(u_i)$ will be a sample from a standard normal distribution. We then have that

$$\log S_T = \log S_0 + (r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T} N^{-1}(u_i)$$

will provide one simulation of S_T , so to calculate

$$e^{-rT} \hat{E} [S_T^2]$$

we require

$$e^{-rT} \frac{1}{n} \sum_{i=1}^n \left(S_0 \exp\{(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T} N^{-1}(u_i)\} \right)^2$$

which can be written equivalently as

$$\frac{S_0^2}{n} e^{(r-\sigma^2)T} \sum_{i=1}^n \exp \{ 2\sigma\sqrt{T} N^{-1}(u_i) \}.$$

5. In the risk-neutral world the stock has the SDE

$$dS_t = rS_t dt + \sigma S_t dB_t.$$

The bond price process grows at the risk-free rate, thus $R_t = R_0 e^{rt}$. The stock price process relative to the riskless bond, Y_t say, satisfies $Y_t = S_t/R_t = S_t e^{-rt}/R_0$. Using Itô's lemma on the function $Y(S, t) = S e^{-rt}/R_0$ we have

$$\frac{\partial Y}{\partial t} = -r S e^{-rt}/R_0 = -rY; \quad \frac{\partial Y}{\partial S} = e^{-rt}/R_0, \quad \frac{\partial^2 Y}{\partial S^2} = 0.$$

So, Itô's lemma gives

$$dY_t = \frac{\partial Y}{\partial S} dS_t + \frac{\partial Y}{\partial t} dt + \frac{1}{2} \frac{\partial^2 Y}{\partial S^2} b^2(S_t, t) dt$$

which reduces to

$$dY_t = \frac{1}{R_t} dS_t - rY_t dt + 0 dt$$

Upon using the fact that $dS_t = rS_t dt + \sigma S_t dB_t$ we have

$$\begin{aligned} dY_t &= \frac{1}{R_t} (rS_t dt + \sigma S_t dB_t) - rY_t dt \\ &= rY_t dt + \sigma \frac{S_t}{R_t} dB_t - rY_t dt \\ &= \sigma \frac{S_t}{R_t} dB_t. \end{aligned}$$

Therefore as the dt term in the SDE for Y is zero the process Y has no drift.